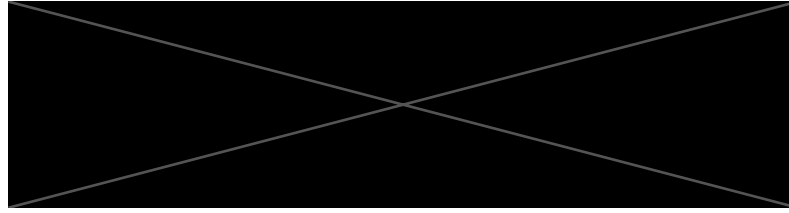


Machine learning Approaches to detect AI-Generated Content



Link to Code Repository:



1 Problem statement and motivation

In the realm of Natural Language Processing (NLP), the journey of development and transformation have been truly remarkable. Evolving from rule-based systems to sophisticated models, NLP has undergone profound changes in recent decades, emerging as an indispensable tool across various domains.

The emergence of ChatGPT stands as a significant milestone, illustrating the unprecedented potential and impact of NLP technologies, pushing in a new era of innovation. Since its launch on November 30, 2022, ChatGPT has garnered a massive user base with over 57 million active users within its first month. Its ability to understand and generate human-like text sets it apart from conventional NLP models, propelling its position as a frontrunner in the field of NLP.

With vast amounts of data and unique architecture, ChatGPT demonstrates versatility across diverse domains and tasks, from customer service, personal assistant, translation, code generation in multiple programming languages, to content creation. Its adaptability has made it indispensable for individuals and organizations, reshaping the traditional way to interact with technology and consume information.

Despite the remarkable capabilities of AI in NLP, it's crucial to acknowledge the challenges and ethical issues associated. The prevalence of concerns such as information bias (Singh and Ramakrishnan (2023)), misinformation and disinformation (Kreps et al. (2022)), and academic integrity issues (Susnjak (2022);Perkins (2023)).

Therefore, it is imperative to comprehend these AI tools, distinguishing between human and AI-generated content patterns to effectively address these ethical concerns. By analyzing data from both human-generated and AI-generated sources

and employing various machine learning models, including Random Forest, Long short-term memory (LSTM), and Bidirectional Encoder Representations from Transformers (BERT), this paper aims to discern the origins of content and understand AI-generated content characteristics, ultimately shaping the future of human-machine interactions.

2 Research hypothesis

In the task of detecting AI-generated text, BERT is expected to outperform Random Forest and LSTM, as indicated by the findings of Theocharopoulos et al. (2023). This superiority can be attributed to BERT's training on extensive text corpora, enabling it to effectively capture deep contextual understanding. Additionally, BERT's bidirectional nature allows it to capture long-range relationships in both left and right directions. Conversely, In contrast, Random Forest and LSTM may struggle to generalize well to unseen data due to the limited training data.

3 Related work and background

The exploration of AI-generated content detection has seen considerable advancement through numerous studies, each offering valuable insights into methodologies and challenges within this domain.

Alamleh et al. (2023) conducted an experiment to assess the ability of machine learning algorithms to distinguish between responses from ChatGPT and computer science students. Their study employed both classical and deep learning models with the TfidfVectorizer feature extractor. Results indicated Random Forest's superior efficacy, outperforming BERT without fine-tuning by 25% in accuracy. Despite the importance of fine-tuning in the BERT model, Random Forest was included in the experiment due to its efficient performance to this particular task.

Similarly, Katib et al. (2023) explored classical and deep learning models, employing TF-IDF,

word embedding, and countvectorizer techniques to distinguish AI-generated text. Their investigation highlighted the TSA-LSTM-RNN model's superior detection efficacy, suggesting the potential of recurrent neural networks in uncovering patterns within AI-generated content.

In a parallel study, Similarly, [Islam et al. \(2023\)](#) undertook a study focusing on content obtained from two diverse sources, Quora and CNN, alongside content paraphrased by AI algorithms. The author proposed the utilization of stopwords as a significant feature to differentiate between human-generated and AI-generated content. Stopwords, being commonly occurring words in a language, were suggested to play a pivotal role in discerning the distinct linguistic patterns as human-generated and AI-generated content tends to exhibit different characteristics in the usage of stopwords. Consequently, this study aims to further explore the effectiveness of utilizing stopwords in detecting AI-generated text. Experiments will compare the performance of models trained with and without stop words as key features.

Two studies, conducted by [Nguyen et al. \(2023\)](#) and [Theocharopoulos et al. \(2023\)](#) are noteworthy. The first study collected human-generated data from news articles, while the second study involved utilizing abstracts from published papers. Both studies employed machine learning-generated datasets by prompting title keywords from human-generated data to generate AI-generated content, leveraging ChatGPT for this purpose.

In their study, [Nguyen et al. \(2023\)](#) employed various feature engineering techniques, including TF-IDF, Named Entity Recognition (NER), N-gram analysis, PCA, and readability scores. Their findings showcased the efficacy of models such as Random Forest and XGBoost in correctly identifying sources. Additionally, they utilized feature importance and SHAP (SHapley Additive exPlanations) analysis to gain insights into the contribution of different features to the detection system, underscoring the utility of explanation frameworks as a valuable tool for discerning differences between models.

[Theocharopoulos et al. \(2023\)](#) conducted data preprocessing by eliminating frequently occurring words like "paper" and "study" and employed a combination of techniques including TF-IDF, NER, and Word2Vec embeddings. Their findings unveiled the superior performance of fine-tuned

BERT and LSTM models with the pretrained embedding model. Given that the data was collected from a single topic related to abstracts. In contrast to their approach, our study encompasses multiple topics and domains. Consequently, removing top frequent words might have an adverse effect on model performance.

[Guo et al. \(2023\)](#) utilized a dataset comprising responses from five distinct domains, employing RoBERTa and logistic regression trained on the GLTR Test-2. The study analyzed the contrasting features between human and AI-generated text, specifically focusing on sentiment and part-of-speech tagging. While the analysis uncovered that humans typically express more negative emotions than AI and display variations in part-of-speech tagging proportions, these variances were not deemed significant. Moreover, the study highlighted RoBERTa's proficiency in the QA detection task, particularly its superior performance in generalizing to unseen data without being influenced by predictable words.

[Mitrović et al. \(2023\)](#) and [Bv et al. \(2024\)](#) employed DistilBert to discern text generated by ChatGPT, focusing on the challenge posed by short text. They highlighted DistilBERT's efficient performance in distinguishing patterns between AI and human-generated text, even when the source was paraphrased from human text or subjected to constraints of short sentence length.

Furthermore, [Silva Barbon and Akabane \(2022\)](#) conducted experiments with various types of BERT models, comparing them across different datasets. Notably, DistilBERT, being 40% smaller and consuming 45% less time compared to the standard BERT model, still manages to retain 97% of its performance, [Sanh et al. \(2019\)](#). This suggests that the size of BERT may not necessarily correlate with its effectiveness. Therefore, this study will experiment with the performance of various types of BERT models to assess their efficiency.

3.1 Accomplishments

- Task 1: Preprocess dataset, by removing noises such as links, and convert to lowercase – Completed
- Task 2: Tokenised the dataset – Completed
- Task 3: Apply Lemmatization – Completed
- Task 4: Build and train random choice

model as baseline on dataset and examine its performance - Completed

- Task 5: Build and train majority class prediction model as baseline on dataset and examine its performance - Completed
- Task 6: Build and train Random Forest with CountVectorizer on dataset and examine its performance - Completed
- Task 7: Explore the influence of stop-words on identifying patterns in AI-generated responses - Completed
- Task 8: Conduct an experiment utilizing CountVectorizer with various n-gram configurations, including 1, 2, 3, 4, and 8 grams - Completed
- Task 9: Build, conduct hyperparameter tuning, and train LSTM model and examine its performance - Completed
- Task 10: Explore the implications of employing Bidirectional LSTM instead of traditional LSTM - Completed
- Task 11: Explore the implications of utilizing GloVe embeddings - Completed
- Task 12: Explore the implications of utilizing Word2Vec embeddings - Completed
- Task 13: Conduct an experiment to determine whether fine-tuning BERT models is significant for optimal performance - Completed
- Task 14: Build and train BERT model using various pretrained models and examine their performance - Completed
- Task 15: Perform in-depth error analysis to figure out what kinds of examples our approach struggles with - Completed
- Task 16: Utilize model interpretability technique - LIME to analyse the importance of each feature and its effect on model predictions - Completed

4 Approach and Methodology

The study is divided into two phases: baseline and main model implementation. In the baseline phase, random guessing and majority prediction models are employed to demonstrate the lowest performance benchmarks. The main models phase utilizes a classical model (Random Forests), a neural network model (LSTMs), and a transformer-based model (BERTs) to explore their distinctiveness, advantages, and weaknesses. Despite their

differences, both phases utilize identical data pre-processing techniques, data partitioning, and evaluation methods, leveraging a confusion matrix and a classification report. Notably, all models are implemented and trained within a "train.ipynb" notebook, with separate sections for each type of model and utility functions stored in a "tools.ipynb" notebook. Model performance evaluation is conducted using "test.ipynb" for Random Forests and LSTMs, while "test-BERT.ipynb" is used for all BERTs. Trained models are saved in the "Models" folder within the same directory, and the source data is maintained in the "source" directory.

Random Forest is an ensemble model that utilizes multiple decision trees, each trained on a randomly selected subset of the dataset, to generate a single result using averaging or voting mechanism. Utilizing ensemble learning, the Random Forest models are expected to effectively discern patterns from the dataset, potentially resulting in superior performance compared to baselines that lack the capacity to learn patterns entirely.

CountVectorizer method is employed with Random Forests to extract features by capturing word frequency. However, relying on word frequency alone may not optimize performance due to noise introduced by commonly used words in the dataset. Consequently, an experiment is conducted to explore the potential enhancement achieved by incorporating TF-IDF with CountVectorizer. This combination aims to reduce noise from common terms while identifying unique characteristics and assessing whether capturing word importance improves the model's efficacy.

Despite its strong capability, Random Forest lacks the ability to capture sequential context effectively. To address this limitation, an experiment on N-gram analysis is conducted, leveraging its potential to capture context within a specified window size. The experiment contains 1, 2, 3, 4, and 8 grams to assess whether employing higher-order N-grams enhances pattern capture.

Following data preparation and feature extraction, the parameter selection process for Random Forests involves utilizing Grid Search with 5-fold cross validation on three key parameters: the number of trees, the maximum depth of each tree, and the minimum number of samples. This approach aims to identify the optimal configuration, mitigating overfitting and maximizing model performance.

In this study, LSTM is explored for its profi-

ciency in capturing long-range dependencies. Unlike Random Forest with N-grams, which are limited to fixed window sizes and treat each sequence of words independently, LSTM can handle variable-length sequences of data and capture semantic relationships within sequential data effectively. Leveraging its superior capabilities, LSTM is expected to outperform both Random Forest and the two baseline models in performance.

Furthermore, to enhance the LSTM model's capabilities, three experiments are conducted, including BiLSTM, BiLSTM with GloVe embedding, and BiLSTM with Word2Vec embedding. These expanded approaches are expected to enhance model performance from the standard LSTM. BiLSTM, by processing sequences in both forward and backward directions simultaneously, offers an enhanced understanding of sequence structure and context. Additionally, the integration of GloVe pre-trained word embedding, learned from extensive text corpora and diverse domains is expected to improve the model's comprehension of semantic relationships and better generalization. Similarly, the utilization of Word2Vec embedding trained on the dataset is expected to provide insights into context-specific patterns, also resulting in enhancing the semantic understanding and generalization capabilities.

After transforming the text data into sequences, where the maximum vocabulary size is set to 10,000, the sequences are padded to ensure uniform length, with a maximum sequence length of 500. The LSTM architecture employed in this study contains an embedding layer, two LSTM layers utilizing hyperbolic tangent function for output and sigmoid function for the recurrent step, along with a dropout rate, and an output layer with sigmoid activation. The loss function is binary crossentropy, and Adam serves as the optimizer. Due to resource constraints, parameter selection for LSTMs adopts a random search approach, prioritizing loss from the validation set. These parameters include embedding dimension, hidden size, dropout rate, and learning rate. Following parameter search, the models are retrained with the optimal parameter configurations. Additionally, to mitigate overfitting and improve model convergence, early stopping and learning rate reduction techniques are utilized, with a primary focus on minimizing validation loss. These strategies help the model to generalize well to unseen data and achieves optimal performance.

Similarly, BiLSTM follows identical architecture and parameter selection procedures, as well as model training techniques as LSTM. The main differences are the integration of bidirectional LSTM layers and the utilization of the Adamax optimizer instead of Adam.

Regarding embedding models, the sequences of data are padded to match the maximum length of sequences within the training data. GloVe embeddings are pretrained from a corpus of 6 billion tokens, resulting in 100 dimensional text encoded vectors for each word, while Word2Vec embeddings are trained specifically on the dataset.

Following the exploration of neural network approaches, transformer-based models, specifically BERTs, are explored. Unlike LSTM and BiLSTM, which process sequences sequentially, BERT's architecture allows it to consider the entire context in both directions simultaneously, regardless of sequence length. Additionally, leveraging its pre-training on large corpora, BERT demonstrates an ability to generalize across diverse domains. These superior capabilities suggest the better understanding the context of each word in a sentence and capturing intricate pattern, leading to enhanced performance. Moreover, the study includes an experiment to evaluate the importance of fine-tuning BERTs compared to freezing weights. To comprehensively assess the performance of BERTs across various architectures and complexities, this study utilizes BERT-Tiny, BERT-Small, BERT-Medium, BERT-Base, and DistilBERT, with all are trained on English uncased text.

Compact versions of BERTs, including BERT-Tiny, BERT-Small, and BERT-Medium are introduced by Google Research (Turc et al., 2019), aimed at reducing size and complexity by decreasing the number of layers and hidden sizes. Meanwhile, DistilBERT is developed by Huggingface researchers Sanh et al. (2019), utilizing distillation technique to learn from larger BERT-Base model.

The BERT preprocessor and classifier are initialized with preset configurations. The preprocessor prepares the input data for the model, setting the sequence length at 256, while the classifier employs the BERT architecture for classification tasks. Given the resource intensive nature of BERTs, iterative parameter selection is impractical for this study. Categorical cross-entropy serves as the loss function, while Adam is chosen as the optimizer. To prevent overfitting, early stopping is implemented,

In this study, a range of libraries and frameworks is employed to serve specific task requirements. NLTK is utilized for data preprocessing and feature engineering tasks such as tokenization, lemmatization, and handling stopwords. The scikit-learn library offers support for data partitioning, CountVectorizer, TF-IDF vectorizer, Grid Search, Random Forest, and evaluation metrics. For deep learning models, Keras provides efficient implementations for training LSTM, covering model architecture, sequence conversion, padding, parameter selection, and model training. Additionally, KerasNLP provides comprehensive support for all BERT models utilized in the study.

This study utilized the Human ChatGPT Comparison Corpus (HC3) dataset, acquired from Hugging Face that introduced by [Guo et al. \(2023\)](#). While the dataset originally includes both English and Chinese language data, this study specifically focuses on English responses. Gathered from multiple sources across diverse domains such as open-domain, psychology, finance, computer science, law, and medicine, this dataset contains questions and their respective answers provided by human experts and ChatGPT. The dataset highlights the complexity of enabling models to learn diverse patterns and generalize effectively across various subject areas.



A bar chart comparing the number of records in Train, Test, and Validation sets for two datasets: Human and Chat-GPT. The y-axis represents the 'Number of records' ranging from 0 to 16,000. The x-axis shows the two datasets. For the Human dataset, the Train set has approximately 16,200 records, while the Test and Validation sets each have approximately 5,400 records. For the Chat-GPT dataset, the Train set also has approximately 16,200 records, with the Test and Validation sets each having approximately 5,400 records. A legend indicates that brown bars represent the Train set, orange bars represent the Test set, and blue bars represent the Validation set.

Dataset	Train	Test	Validation
Human	~16,200	~5,400	~5,400
Chat-GPT	~16,200	~5,400	~5,400

from Wikipedia, as shown in Figure 1. Both human and ChatGPT answers are stored within lists associated with the same questions.

To balance the dataset between human and AI answers, the data was downsampled, resulting in 26,903 records each. Subsequently, 20% of the total data was reserved for validation and testing purposes, with 10,760 records allocated to each

[illegible]

Figure 3: Human's Word Cloud analysis

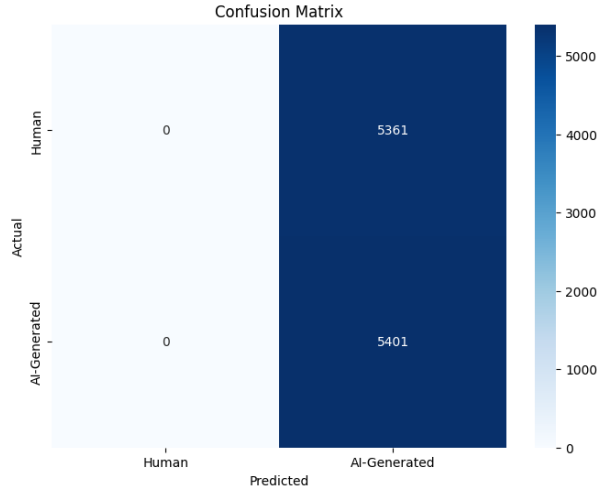


Figure 7: Confusion Matrix: Majority Voting model

The second baseline is the majority vote model, predicted based on the majority label in the dataset, resulting in a lower performance with 50% accuracy and only 33% F1 score, as the data is balanced among classes.

Despite their simplicity and efficiency in construction, these baseline models demonstrate limited effectiveness. In terms of baselines, the random guess and majority vote model provide a benchmark for the lowest performance expectation and represents a simple approach for classification. These baselines are useful for evaluating the performance of more complex models and assessing their effectiveness compared to simple strategies.

7 Results, error analysis

Following preprocessing, this study explores into experimentation with three main models: Random Forests, LSTMs, and BERTs. Through rigorous testing, the study achieves an impressive 99% accuracy and F1 score.

7.1 Random Forest

The initial experiment starts with the Random Forest model, exploring various configurations such as N-grams analysis and the efficacy of combining TF-IDF with CountVectorizer.

Based on the experimental findings shown in Table 3, it is observed that Random Forest with unigram achieved the highest effectiveness compared to higher order N-grams. This result suggests that individual features effectively represent the data, while higher order N-grams may introduce noise and irrelevant information, leading to reduced performance.

Table 3: Performance metrics for N-gram analysis

N-gram	Accuracy	F1 Score
1-gram	0.94	0.94
2-gram	0.90	0.90
3-gram	0.86	0.86
4-gram	0.80	0.79
8-gram	0.62	0.56

Table 4: Performance metrics reflecting the efficacy of TF-IDF

Approach	Accuracy	F1 Score
CountVectorizer	0.94	0.94
CountVectorizer + TF-IDF	0.93	0.93

Moreover, The result in Table 4 indicates that employing CountVectorizer alone outperforms combining CountVectorizer with TF-IDF. This underscores the importance of word frequency over their uniqueness within this dataset. For further details of experimental results and analysis, please refer to the supplementary document. Throughout experiments, the Random Forest model with CountVectorizer and unigram achieves a remarkable 94% accuracy and F1 score, significantly outperforming baseline models.

Error Analysis

- Misclassified from Human to AI: “I know this is ELI5 , but I ’d like to add that , as with everything in English, there is an exception...”.
- Misclassified from AI to Human: “Digg was a website that allowed users to share and discover interesting news and content from around the internet. It was very popular in the late 2000s and early 2010s...”.

The examples illustrate a different pattern between human and AI generated text. Human text indicates a casual, conversational tone, while AI-generated text tends to be more formal and information-oriented. However, these distinctions also reveal the limitations of the model. Depending solely on word frequency and occurrence, the model fails to notice the essential language elements such as understanding sequence and context. Consequently, it faces challenges in accurately categorizing text that differs from its training data characteristics.

The LIME chart presented in Figure 9 illustrates the model’s preference for frequently occurring words. Notably, it demonstrates that common

words hold more significance in AI-generated text, compared with human-written text.

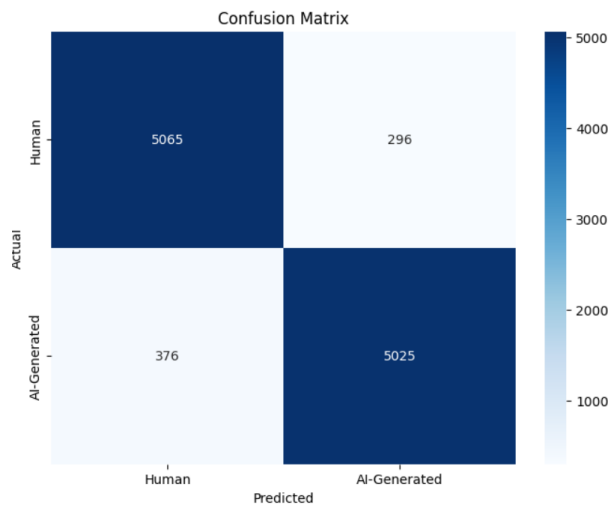


Figure 8: Confusion Matrix: Random Forest

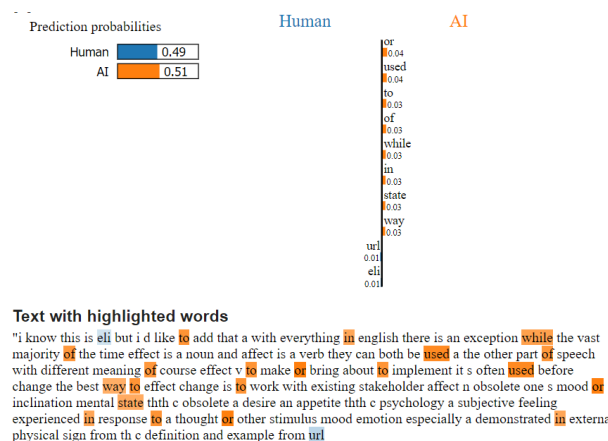


Figure 9: LIME based on Random Forest

7.2 LSTM

The results presented in Table 5 highlight that LSTMs exhibit superior performance compared to Random Forests, achieving 98% accuracy and F1 score. This is attributed to the sequential nature of LSTM models, which enables them to effectively capture contextual relationships within the data.

Additionally, both BiLSTM and BiLSTM with GloVe embeddings outperform other LSTM variants in this experiment. The enhanced performance of BiLSTM models can be attributed to their ability to process sequences in both forward and backward directions, facilitating a deeper understanding of relationships. However, BiLSTM with Word2Vec embedding demonstrates relatively low performance, possibly due to limitations in capturing

semantic relationships within the Word2Vec embeddings.

Although both BiLSTM and BiLSTM with GloVe embeddings achieved equal accuracy and F1 scores, the slight enhancement in overall performance with GloVe embeddings can be attributed to their pre-training on large corpora covering diverse topics (Refer to the supplementary document). These results underscore the efficacy of sequential models and the extended advantage of leveraging pre-trained word embeddings to boost model performance.

Table 5: Performance metrics for various LSTMs

N-gram	Accuracy	F1 Score
LSTM	0.96	0.96
BiLSTM	0.98	0.98
BiLSTM+GloVe	0.98	0.98
BiLSTM+Word2Vec	0.84	0.84

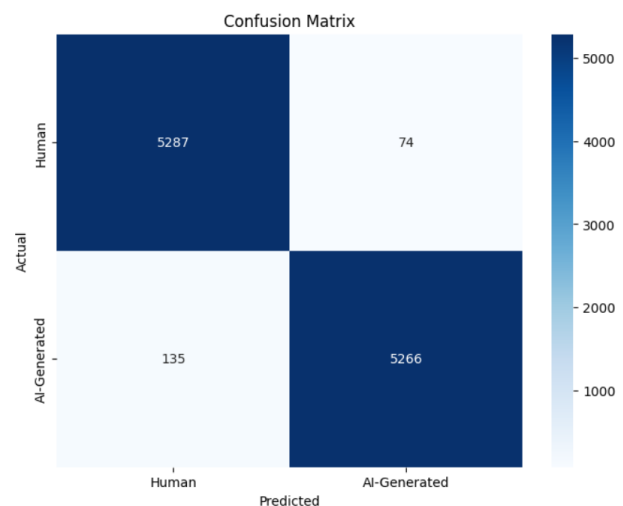


Figure 10: Confusion Matrix: BiLSTM with GloVe

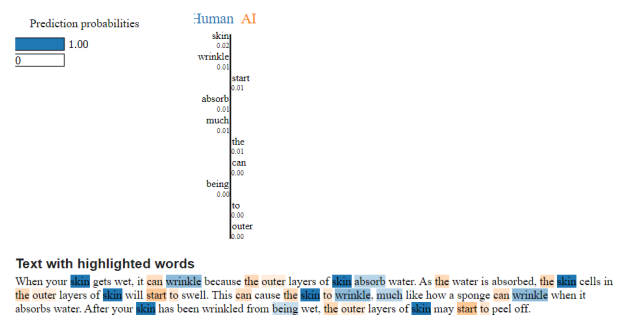


Figure 11: LIME based on BiLSTM

Error Analysis

- Misclassified from Human to AI: "Octane is

a rating we give fuels that express their resistance to detonation under pressure. In car engines we do not want fuels to ignite prematurely. We want them to wait until the proper moment. But if a fuel is too sensitive to pressure, it may ignite as it is being squeezed by the piston before reaching the proper moment. This is known as detonation. ”

- Misclassified from AI to Human: “When your skin gets wet, it can wrinkle because the outer layers of skin absorb water. As the water is absorbed, the skin cells in the outer layers of skin will start to swell. This can cause the skin to wrinkle, much like how a sponge can wrinkle when it absorbs water. After your skin has been wrinkled from being wet, the outer layers of skin may start to peel off.”

From the examples, despite the similar task asking to describe the concepts, both answers from humans and AI are in the similar lengths, the written style is completely different. AI-generated text is more straightforward with narrative information and clear logic, providing step-by-step explanation. However, BiLSTM cannot handle this difference, as it struggles to capture long-term dependencies in long answers. The task to differentiate effectively AI-generated text requires understanding the structure in long distance, but BiLSTM process sequential and it can lose context due to long distance. Also, lack of vocabulary from the training set can cause difficulty in understanding the pattern accurately.

7.3 BERT

The studies conducted by [Alamleh et al. \(2023\)](#) and [Theocharopoulos et al. \(2023\)](#) highlight the significance of fine-tuning BERT models compared to freezing weights during training. As indicated by the experimental results in Table 6, fine-tuning BERT models leads to a remarkable increase in accuracy, achieving 99% compared to 80% when weights are fixed.

Table 6: Performance metrics reflecting the effectiveness of fine-tuned BERT

	Accuracy	F1 Score
Fixed weights	0.80	0.79
Fine-tuning	0.99	0.99

In this study, various versions of BERT models are examined to evaluate their performance across

different architectures and complexities. The results presented in Table 7 highlight the effectiveness of fine-tuned BERT models across different architectures, ranging from BERT-Tiny to BERT-Base. Notably, even the smallest BERT model with only 2 layers demonstrates comparable overall performance to larger model like BERT-Base, surpassing Random Forests and LSTMs.

Table 7: Performance metrics for various BERTs

BERTs	Accuracy	F1 Score
DistilBERT	0.99	0.99
BERT-Base	0.99	0.99
BERT-Medium	0.99	0.99
BERT-Small	0.99	0.99
BERT-Tiny	0.99	0.99

However, slight variations in performance are captured among different BERT types (Refer to supplementary document), which align with the complexity of the models. BERT-Base achieves the highest performance, while BERT-Tiny exhibits the lowest. Despite the high accuracy of BERT-Base, it comes with increased model complexity and resource requirements compared to smaller variants. This indicates that intermediate complexity BERT models like BERT-Medium and DistilBERT might provide a better trade-off between efficiency and efficacy. Despite a slight decrease in performance, there is a significant decrease in time consumption.

Error Analysis

- Misclassified from Human to AI: “Humboldt is a city situated along the Neosho River in the southwest part of Allen County , located in southeast Kansas , in the Central United States”
- Misclassified from AI to Human: “Sue Lyon was 14 years old when she starred as Dolores "Lolita" Haze in the film "Lolita" (1962).”

While the BERT model demonstrates high performance in identifying AI-generated responses, its effectiveness is constrained when dealing with short answers. As illustrated in the examples, the lack of context in these responses can result in misclassifications.

Moreover, sentences with complex structures and ideas cause challenges for the BERT model, also leading to incorrect classifications. An example of such a sentence is: "Imagine if you will that

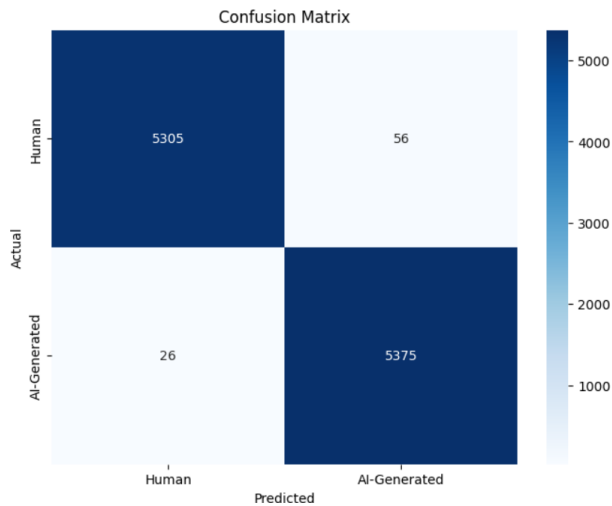


Figure 12: Confusion Matrix: BERT-Base

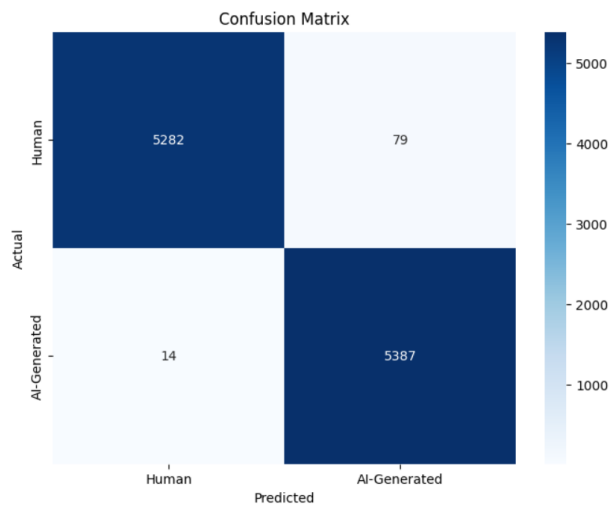


Figure 13: Confusion Matrix: BERT-Medium

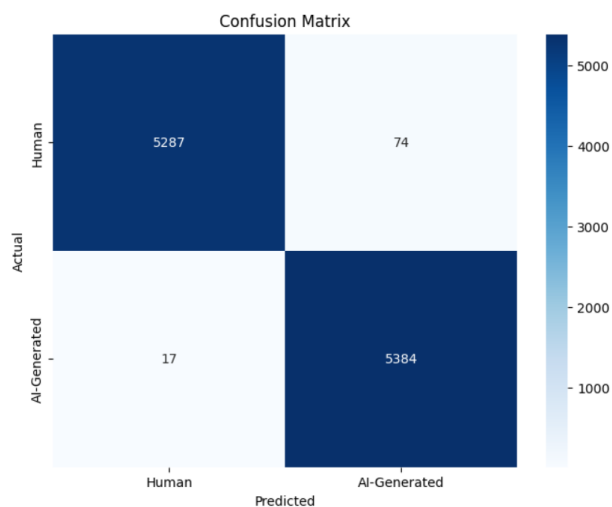


Figure 14: Confusion Matrix: DistilBERT

you have a group of people called A , B , C , D and E. All of these people are then matched up randomly so in giving gifts you could have Santal Receiver A | C B | D C | E D | A E | B This means that everyone is paired up with both someone to give them a gift and someone to give a gift to ...".

8 Lessons learned and conclusions

In conclusion, this study aimed to differentiate the patterns between human and AI responses using Random Forests, LSTMs, and BERTs, compared to baseline models, including random guessing and majority vote. The experimental results revealed the effectiveness of employing stopwords as key features, leading to a 6% increase in accuracy. In particular, the Random Forest model performed best with unigram and CountVectorizer, achieving 94% accuracy, a significant improvement over higher order N-grams and the combination of CountVectorizer with TF-IDF. LSTMs, with their ability to capture sequential information, further enhanced performance to 96% accuracy. Moreover, leveraging bidirectional architecture, BiLSTM achieved a performance boost to 98% accuracy with both custom and pretrained word embeddings like GloVe.

Further exploration into transformer-based models, specifically BERTs, demonstrated an improvement in discerning patterns, reaching 99% accuracy across all variations, including BERT-Base, BERT-Medium, BERT-Small, BERT-Tiny, and DistilBERT. The study highlighted the importance of fine-tuning BERT models, with notable performance improvements from 80% to 99% accuracy.

However, considering the high complexity and resource consumption of BERT-Base, it may not be the most preferable option. Instead, BERT-Medium offer a better balance of performance and efficiency.

In future work, it would be valuable to explore further into feature extraction techniques such as Named Entity Recognition (NER) and Part-of-Speech (POS) tagging, as these methods have the potential to enhance model performance. Additionally, conducting extensive hyperparameter tuning for BERT models could result in significant improvements in accuracy and robustness.

In conclusion, the results of this study offer great potential for improving the field of AI-generated text detection, aiming to enhance transparency and reduce ethical concerns like misinformation and

information bias. However, it is still crucial to continue refining these methods to ensure the ethical and responsible use of AI in text analysis.

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